

Direct Carboxylative Reactions for the Transformation of Carbon Dioxide into Carboxylic Acids and Derivatives

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Abstract: Carbon dioxide is the ideal one-carbon source for organic synthesis because of its abundance, lack of toxicity, and potential as a renewable resource, but is limited by its high stability and low reactivity. The carboxylation of carbon nucleophiles with carbon dioxide to form new carbon–carbon bonds is therefore an attractive method for the synthesis of carboxylic acids and derivatives, which are in turn valuable organic products. Thus, the designing of mild methods to catalytically activate carbon dioxide and form carbon–carbon bonds is a challenge that is of both academic and practical importance. This review is focused on the direct carboxylative reaction for the transformation of carbon dioxide into carboxylic acids and derivatives from the carboxylation of carbon nucleophiles, reductive hydrocarboxylation of unsaturated compounds, carboxylation via oxidative cycloaddition, carboxylation of carbon–hydrogen bonds, and electrochemical carboxylation.

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Key words: carbon dioxide, carboxylic acid, carboxylic acid derivative, carboxylative reaction, renewable resources

1 Introduction

The increase in atmospheric carbon dioxide (CO_2) is linked to climate changes, hence there is an urgent need to reduce the accumulation of carbon dioxide in the atmosphere. Meanwhile, carbon dioxide is not only a greenhouse gas but is also the greatest carbon resource in the world. In recent years, much attention has been paid to the chemical fixation of carbon dioxide to synthesize useful

intermediates and chemicals.¹ Carbon dioxide is an ideal carbon source owing to its abundance, low cost, lack of toxicity, and good potential as a renewable source.² However, it is not easy to activate such a thermodynamically and kinetically stable material, and a catalyst must therefore play an important role in the development of useful transformations with this attractive and environmentally friendly raw material. Thus, designing mild methods to catalytically activate carbon dioxide and to form carbon–carbon bonds is a challenge that is of both academic and practical importance.

Carboxylic acids and derivatives are some of the most important types of compounds in medicinal chemistry and in fine-chemicals synthesis. Although there are many well-established protocols for the preparation of carboxylic acids and derivatives, the direct carboxylation of carbon nucleophiles using carbon dioxide as the electrophile is the most attractive and straightforward method.³ The formation of a stable carbon–carbon bond is desired for carbon dioxide fixation and remains the most challenging aspect thus far. Typically, the insertion of carbon dioxide into metal–carbon bonds of various organometallic reagents with or without catalysts has been well documented.⁴ Widespread use of these methods is limited by the synthesis of organometallic reagents as precursors and the restricted substrate scope.

The transition-metal-catalyzed carboxylation of various nucleophiles or unsaturated hydrocarbons using carbon dioxide as a carboxylative agent is an attractive route to produce carboxylic acids and derivatives. In most of these reactions, a transition-metal catalyst plays an important role to help the reaction proceed smoothly in an efficient and selective manner under mild conditions. In this review, we focus on the direct carboxylative reaction for the transformation of carbon dioxide into carboxylic acids and derivatives from the carboxylation of carbon nucleophiles, reductive hydrocarboxylation of unsaturated compounds, carboxylation via oxidative cycloaddition, carboxylation of carbon–hydrogen bonds, and electrochemical carboxylation.

2 Carboxylation of Carbon Nucleophiles

Carbon–carbon bond-forming reactions are the result of a wide variety of organic transformations employing carbon dioxide. Since carbon dioxide is an inert electrophile, highly nucleophilic reagents are essential for the efficient

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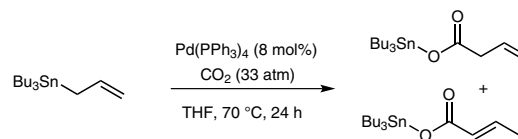
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activation of carbon dioxide. The best-known methods for realizing the carbon–carbon bond-forming reaction with carbon dioxide are the reactions with Grignard and organolithium reagents, and these highly nucleophilic reagents react with carbon dioxide without a catalyst. However, in these reactions, many useful functional groups such as ketones, esters, and nitriles are not tolerated. Thus, in order to develop transformations compatible with various functionalities, less nucleophilic substrates must be employed, in which case catalysts are indispensable for successful carbon–carbon bond formation reactions. Transition-metal complexes were confirmed to directly catalyze the carboxylation of less reactive carbon nucleophiles, and others, with carbon dioxide to form valuable carboxylic acids and their derivatives under milder reaction conditions.⁵ In addition to active organolithium and Grignard reagents, other organometallic reagents, including organotin, organoboron, and organozinc reagents, have been used successfully in the carboxylation with carbon dioxide catalyzed by transition-metal complexes.

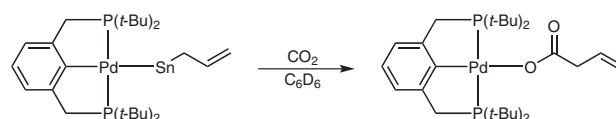
2.1 Carboxylation of Organotin Reagents

Nicholas and Shi reported the first examples of palladium-catalyzed insertion of carbon dioxide into unreactive tin–carbon bonds in 1997.⁶ The relatively active allyltributyltin does not react with carbon dioxide even at 70 °C and high pressure (33 atm) for 24 hours, but the reaction occurred smoothly in the presence of a catalytic amount of Pd(PPh₃)₄ or Pd(PBu₃)₄ (8 mol%), providing the carbox-

ylate mixture in a quantitative yield (Scheme 1). In 2007, Wendt and Johansson reported another example in which the complex {2,6-bis[(di-*tert*-butylphosphino)methyl]phenyl}allylpalladium [PCP(*t*-Bu)Pd-allyl] reacted with carbon dioxide in a very fast insertion reaction to give the corresponding butenoate complex (Scheme 2).⁷



Scheme 1



Scheme 2

In 2011, a family of well-defined (η^3 -allyl)Pd(L)-(carboxylate) ($\text{L} = \text{R}_3\text{P}$ or NHC) complexes were found by Wu and Hazari to be efficient catalysts for the catalytic carboxylation of allylstannanes into allylcarboxylates using carbon dioxide.⁸ The substrate scope of this reaction included both substituted allylstannanes and allylboranes (Scheme 3).

Biographical sketches



Xiaohua Cai was born in 1966. He graduated from Shaoyang Technology University, P. R. of China, in 1988. He received his M.S. in chemistry from Xi'an Modern Chemistry Re-

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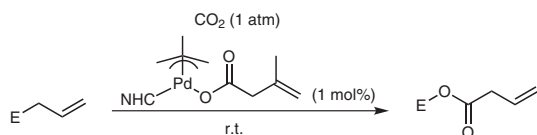
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Bing Xie was born in 1963. He is a senior professor at Guizhou Minzu University, Guiyang (P. R. of China). He graduated from South-

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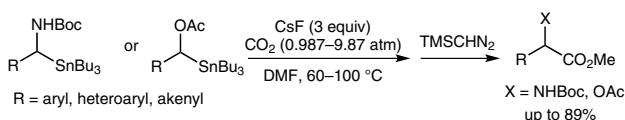
are focused on the synthesis of heterocyclic compounds with prospective biological and pharmaceutical activities.



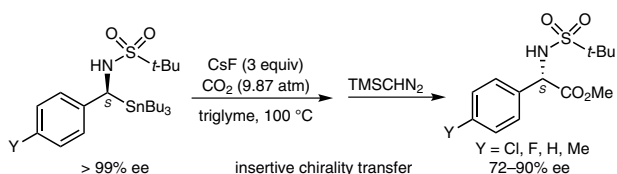
E = Me₃Sn, (*n*-Bu)₃Sn, (RO)₂B
 NHC = 1,3-bis(2,6-diisopropylphenyl)-1,3-dihydro-2*H*-imidazol-2-ylidene

Scheme 3

Recently, Mita, Sato and co-workers developed the incorporation reaction of carbon dioxide with *N*-Boc- α -amido and α -acetoxy stannanes using cesium fluoride as a mild tin activator (Scheme 4).⁹ Monoprotected α -amido stannanes could be used, and the corresponding arylglycine derivatives were obtained in moderate to high yields under 1 MPa (9.87 atm) of carbon dioxide pressure. α -Acetoxy stannanes also underwent carboxylation to afford mandelic acid derivatives in excellent yields under ambient carbon dioxide pressure. Both transformations enabled the synthesis of α -tertiary and α -quaternary carboxylic acid derivatives. In addition, the chirality of (*S*)-*N*-*tert*-butylsulfonyl- α -amido stannanes was transferred with up to 90% inversion of configuration at 100 °C (Scheme 5).



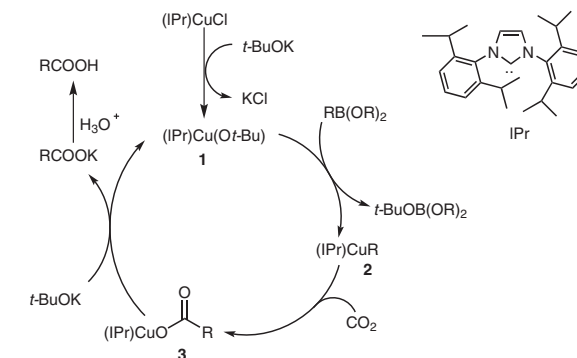
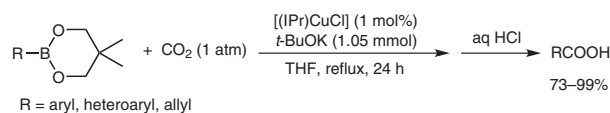
Scheme 4



Scheme 5

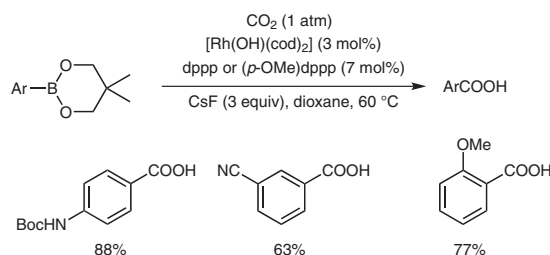
2.2 Carboxylation of Organoboron Reagents

In 2008, Hou and co-workers demonstrated that the combination of [(IPr)CuCl] with potassium *tert*-butoxide could serve as an excellent catalyst system for the carboxylation of aryl- and alkenylboronic esters with carbon dioxide, leading to selective formation of various functionalized carboxylic acid derivatives (Scheme 6).¹⁰ Some key reaction intermediates, such as the copper(I) alkoxide complex (IPr)Cu(O*t*-Bu) (**1**), the alkyl or aryl complex **2**, and the carboxylate complex **3**, have been isolated and structurally characterized, and have been confirmed to be active species in this catalytic process.



Scheme 6

The preparative aspects of using carbon dioxide in organic synthesis as a one-carbon source undoubtedly represent a key strategy for the development of greener chemical processes. In 2009, substantial advances in the field of carboxylation of organometallic reagents allowed the use of less-reactive nucleophilic components, such as boron or zinc reagents.¹¹ These methods were distinguished by their wide scope and functional-group tolerance and therefore might importantly emerge as competitive and convenient methods for the synthesis of carboxylic acid derivatives (Scheme 7 and Scheme 8).

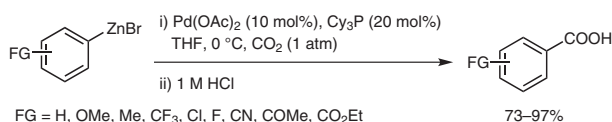


Scheme 7

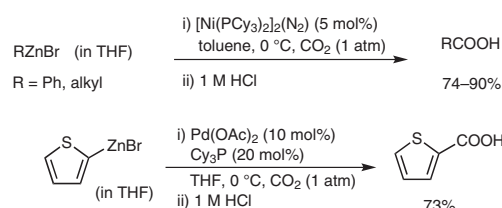
In 2011, Hou and co-workers demonstrated that the combination of [(IPr)CuCl] with methoxylithium could serve as an excellent catalyst system for the carboxylation of alkylboranes with carbon dioxide, leading to the efficient synthesis of various functionalized carboxylic acids (Scheme 9).¹² The isolation of the copper methoxide-alkylborane adduct **4** and its carbon dioxide insertion product **5** provided important information for understanding the mechanistic details. These results might also offer new insight into other catalytic cross-coupling reactions involving metal/organoboron compounds; in particular reactions for the transmetalation of organoboron compounds.

2.3 Carboxylation of Organozinc Reagents

Organozinc reagents are widely used carbon nucleophiles that show less reactivity than organolithium and Grignard reagents and cannot be directly carboxylated by carbon dioxide in the absence of a metal catalyst. However, the exploration of direct carboxylation of organozinc reagents via transition-metal catalysis has been given more attention in recent years. In 2008, Yeung and Dong developed a novel catalytic strategy for the cross-coupling of organozinc reagents with carbon dioxide using nickel and palladium complexes as catalysts in good yield, and palladium(II) acetate was shown to be a convenient catalyst precursor for carbon dioxide activation in these reaction (Scheme 12 and Scheme 13).¹⁶ In the same year, an efficient nickel catalyst system for the carboxylation of organozinc reagents with carbon dioxide under very mild conditions was developed by Yorimitsu, Oshima and co-workers.¹⁷ The catalyst system complemented the conventional methods and enabled the direct synthesis of various saturated carboxylic acid derivatives from the corresponding alkylzinc reagents and carbon dioxide under mild conditions (Table 2).

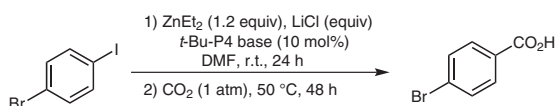


Scheme 12



Scheme 13

In 2009, Kobayashi and Kondo reported an efficient process for the carboxylation of functionalized organozinc reagents with carbon dioxide under transition-metal-free conditions with *N,N*-dimethylformamide as solvent in the presence of lithium chloride (Table 3).⁴ Using this method, *p*-bromiodobenzene was selectively monozincated, and the bromophenylzinc was carboxylated smoothly to give *p*-bromobenzoic acid (Scheme 14).



Scheme 14

In 2010, Knochel and co-workers reported on a variety of functionalized organozinc reagents that underwent smooth addition with carbon dioxide in the presence of

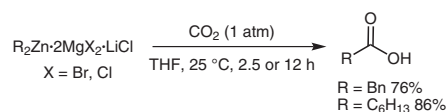
Table 2 Carboxylation of Organozinc Reagents

R	Isolated yield (%) of acid	Isolated yield (%) of ester
<i>n</i> -C ₆ H ₁₃	62	70
PhCH ₂ CH ₂	60	74
Me ₂ C=CH(CH ₂) ₂ CH(Me)CH ₂ CH ₂	50	69
<i>c</i> -C ₆ H ₁₁	62	70
BnO(CH ₂) ₆	60	69
<i>t</i> -BuPh ₂ SiO(CH ₂) ₆	41	51
4-ClC ₆ H ₄ CH ₂ O(CH ₂) ₆	68	69
PhCO ₂ (CH ₂) ₆	67	77
HC(OCH ₂ CH ₂ O)(CH ₂) ₅	—	79
PhC(OCH ₂ CH ₂ O)(CH ₂) ₅	36	50

Table 3 Carboxylation of Functionalized Organozinc Reagents

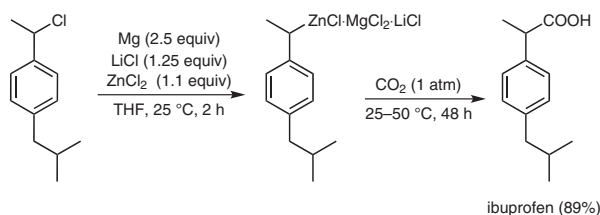
R	Isolated yield (%)
4-MeOC ₆ H ₄	73
4-MeC ₆ H ₄	89
3-MeC ₆ H ₄	78
2-MeC ₆ H ₄	56
4-MeO ₂ CC ₆ H ₄	53
4-ClOC ₆ H ₄	69
4-BrC ₆ H ₄	98
4-IC ₆ H ₄	70
1-naphthyl	100
EtO ₂ C(CH ₂) ₃	66
1-styryl	97

stoichiometric amounts of magnesium chloride at ambient temperature to yield carbonyl compounds (Scheme 15).¹⁸



Scheme 15

The direct reaction with carbon dioxide opens an expeditious route to phenylacetic acid derivatives, as demonstrated in a short synthesis of ibuprofen (Scheme 16).

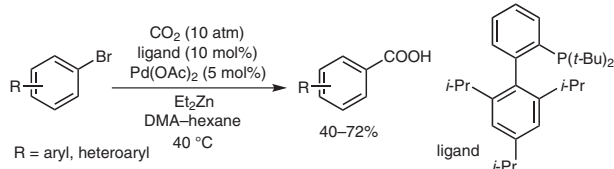


Scheme 16

2.4 Carboxylation of Other Nucleophiles

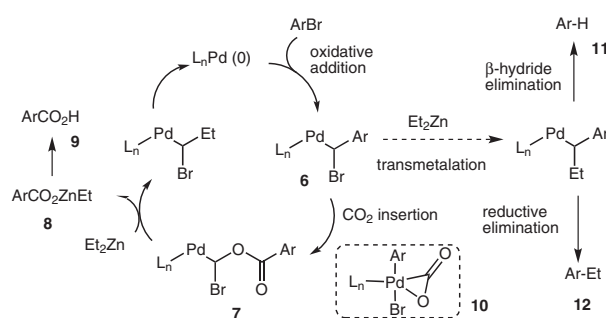
One of the earliest examples dealing with the carboxylation of arylsodium species was reported by Wanklyn in 1858.²⁰ Later, De Pasquale and Tamborski²¹ and Ebert et al.²² developed the carboxylation of arylcopper reagents using carbon dioxide. In 2008, Anastas and co-workers developed the silver(I)-catalyzed reaction of phenylacetylene, carbon dioxide and 3-bromo-1-phenylprop-1-yne, which afforded aryl-naphthalene lactones in poor to moderate yields.²³

In 2009, Correa and Martín developed a novel palladium catalyst system for the carboxylation of aryl bromides with carbon dioxide (Scheme 17).²⁴ There was no need to prepare the corresponding organometallic intermediates, thus constituting an additional advantage in practical and economic terms. This transformation is a straightforward alternative for the synthesis of benzoic acids using carbon dioxide as the sole source of carbon. The proposed catalytic cycle (Scheme 18) implies a challenging carbon dioxide insertion into the palladium–aryl bond of an initially formed complex **6** to yield **7**. Subsequently, transmetalation with diethylzinc would deliver the zinc carboxylate **8**, with concomitant release of **9**, which ultimately would lead to the regeneration of the catalytic palladium(0)–ligand species. At present, the intermediacy of palladium(IV) species **9**, which would subsequently undergo reductive elimination to afford **7**, cannot be excluded. In full accordance with the mechanistic proposal, the formation of **11** and **12** by competitive transmetalation of diethylzinc with intermediate **6** followed by either β -hydride elimination or reductive elimination, respectively, can be rationalized.



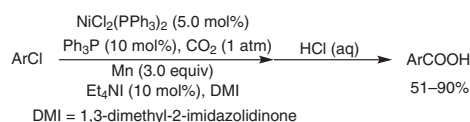
Scheme 17

Recently, Tsuji and co-workers developed a nickel-catalyzed, highly efficient carboxylation of aryl and vinyl chlorides employing carbon dioxide.²⁵ The reactions pro-

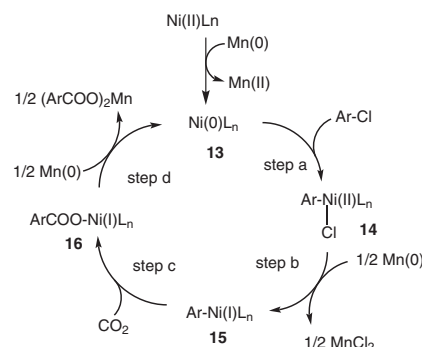


Scheme 18

ceeded under a carbon dioxide pressure of one atmosphere at room temperature in the presence of a nickel catalyst and manganese powder as a reducing agent (Scheme 19). Various aryl chlorides were converted into the corresponding carboxylic acid in good to high yields. Furthermore, vinyl chlorides were successfully carboxylated with carbon dioxide. Mechanistic studies suggested that a nickel(I) species was involved in the catalytic cycle with the aid of the manganese and tetraethylammonium iodide system as an efficient reducing agent. A possible catalytic cycle for the nickel-catalyzed carboxylation of aryl chlorides with carbon dioxide was proposed, as shown in Scheme 15. First, the nickel(II) complex must be reduced to nickel(0) species **13**. Then, oxidative addition of the aryl chloride takes place to give the nickel(II) intermediate **14** (step a). As suggested by the stoichiometric reaction in Scheme 20, nickel(II) would be reduced by the manganese and tetraethylammonium iodide system to afford nickel(I) intermediate **15** (step b). The generation of nickel(I) species was observed in electrochemical reactions. Then, nucleophilic nickel(I) **15** reacts with carbon dioxide to give the carboxylatonic nickel intermediate **16** (step c). Finally, reduction of **16** by manganese gives the corresponding manganese carboxylate, and the nickel(0) catalyst species is regenerated (step d).



Scheme 19



Scheme 20

In 2012, the Kobayashi research group reported that they had obtained 4-substituted 3,4-dihydro-3-methoxyisocoumarins – a new type of isocoumarin (1*H*-isochromen-1-one, 1*H*-2-benzopyran-1-one) – by a one-pot process from α -substituted 2-bromo- β -methoxystyrenes.²⁶ Lithium 2-[1-aryl(or methyl)-2-methoxyethenyl]benzoates were conveniently generated via bromine–lithium exchange between the styrene and butyllithium, followed by the action of carbon dioxide on the resulting α -substituted 2-lithio- β -methoxystyrenes. Upon treatment with concentrated hydrochloric acid at room temperature, these lithium benzoates underwent lactonization to provide the desired 3,4-dihydroisocoumarins in relatively good yields (Table 4).

Table 4 Production of 3,4-Dihydroisocoumarins

R ¹	R ²	Yield (%)
H	Ph	70
H	3-MeC ₆ H ₄	71
H	4-MeC ₆ H ₄	73
H	4- <i>t</i> -BuC ₆ H ₄	60
H	4-ClC ₆ H ₄	72
H	3-MeOC ₆ H ₄	70
H	4-MeOC ₆ H ₄	0
H	4-PhC ₆ H ₄	0
MeO	Me	56
MeO	Ph	60

Recently, Martin and co-workers described a direct and efficient nickel-catalyzed carboxylation of primary, secondary, and tertiary benzyl halides with carbon dioxide via C(sp³)–halide activation under mild conditions (atmospheric carbon dioxide pressure) at room temperature (Table 5 and Table 6).²⁷ The transformation presented a straightforward alternative to other existing methodologies that required the preparation of well-defined and sensitive organometallic reagents. The authors tentatively proposed a catalytic cycle consisting of an initial reduction of the nickel(II) precatalyst promoted by zinc followed by oxidative addition to the corresponding alkyl halide, thus delivering η^3 - and η^1 -bound nickel complexes **18** and **19** (Scheme 21).²⁸ It is expected that **18** and **19** would indeed be in rapid equilibrium as suggested by Jamison in a 2011 disclosure.²⁹ Zinc would then mediate the generation of nickel(I) species **20**, setting the stage for carbon dioxide insertion en route to **21**. A final treatment with zinc would regenerate the active Ni(0)L₂ species **17** and provide zinc carboxylate **22**, which upon hydrolytic workup would deliver the expected product. While the

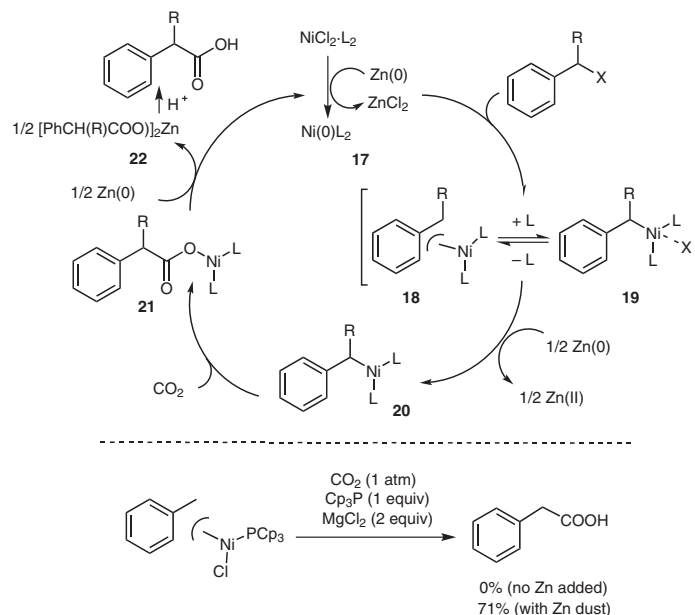
role of magnesium chloride remains to be elucidated, at that time they speculated that this additive might act as either a Lewis acid to facilitate carbon dioxide insertion or as an activator of zinc dust.³⁰ Gratifyingly, the nickel complex did not undergo carboxylation in the absence of zinc but instead afforded phenylacetic acid in 71% yield in the presence of zinc dust.

Table 5 Nickel-Catalyzed Carboxylation of Primary Benzyl Halides

R	X	Yield (%)
4- <i>t</i> -BuC ₆ H ₄	Cl	70
Ph	Cl	67
Ph	Br	69
4-MeOC ₆ H ₄	Cl	69
4-F ₃ CC ₆ H ₄	Cl	70
4-FC ₆ H ₄	Cl	70
2-ClC ₆ H ₄	Cl	61
3-MeO ₂ CC ₆ H ₄	Cl	79
3-TIPSOC ₆ H ₄	Cl	61
2-naphthyl	Cl	72
2-naphthyl	Br	59
4-AcC ₆ H ₄	Cl	49
2,4,6-Me ₃ C ₆ H ₂	Cl	70
2,4,6-Me ₃ C ₆ H ₂	Br	49
4-H ₂ C=CHC ₆ H ₄	Cl	67
PhC $\equiv$ C	Cl	29

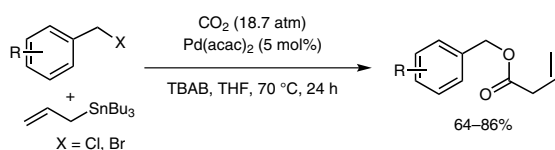
Table 6 Nickel-Catalyzed Carboxylation of Secondary and Tertiary Benzyl Halide

R ¹	R ²	R ³	Yield (%)
Ph	Me	H	58
Ph	Ph	H	47
4-F ₃ CC ₆ H ₄	Me	H	61
4- <i>i</i> -BuC ₆ H ₄	Me	H	59
2,3-dihydro-1 <i>H</i> -indenyl			51
9 <i>H</i> -fluorenyl			60
Ph	Ph	Ph	55



Scheme 21

Recently, Feng, Bao and co-workers reported that palladium-catalyzed carboxylative coupling of benzyl chlorides with allyltributylstannane was successfully conducted to produce benzyl but-3-enoates in satisfactory to good yields (Scheme 22).³¹ The carboxylative coupling reaction occurred smoothly under mild conditions in the presence of palladium nanoparticles in tetrahydrofuran.



Scheme 22

In general, transition metals can play a critical role in the carboxylation of carbon nucleophiles with carbon dioxide. Complexes of transition metals with various ligands can enhance the nucleophilicity of organometallic reagents by tuning the electronic properties and steric circumstances of the metal center, and form the corresponding products in higher yields under milder reaction conditions.

3 Reductive Hydrocarboxylation of Unsaturated Compounds

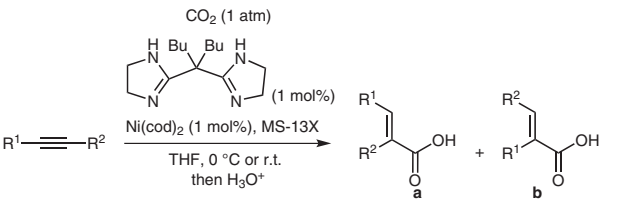
The catalytic generation of nucleophilic organometallic species from easily available unsaturated hydrocarbons is highly desirable from the standpoint of atom-economy in synthetic methodology. Recent breakthroughs in the catalytic hydrocarboxylation of styrene³² and single alkynes^{33,34} with carbon dioxide have extended beyond the limitation of substrates with an extensive π system. The catalytic carbon dioxide hydrocarboxylation of these

simple unsaturated substrates is a significant advance and will prove to be a powerful synthetic approach.³⁵

Nickel may effectively promote the reductive hydrocarboxylation of unsaturated compounds and produce carboxylic acids and derivatives. In the 1980s, Hoberg and co-workers reported that nickel(0) with 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) as ligand-promoted carbon–carbon double-bond formation between alkenes and carbon dioxide.³⁶ Subsequently, insertion reactions of carbon dioxide into organic molecules employing a catalytic amount of nickel catalyst were also accomplished by Mori and co-workers.³⁷

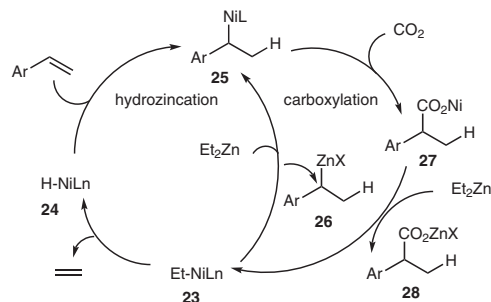
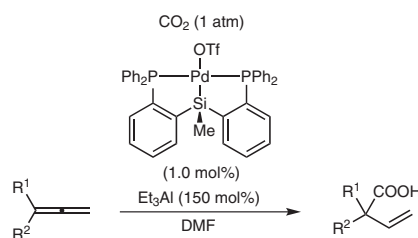
In 2004, Iwasawa and co-workers introduced bidentate amidine ligands in transition-metal-mediated carbon–carbon bond-forming reactions, and applied these bis(amidine)s in the nickel(0)-mediated coupling of carbon dioxide with unsaturated hydrocarbons. Good yields and regioselectivities were achieved when 2,2'-(dibutylmethylene)bis(imidazoline) was employed (Table 7).³⁸ In particular, high regioselectivity was noted for the carboxylation of aryl-substituted internal alkynes. It was also shown that the regioselectivity of this coupling could be controllable by the appropriate choice of bis(amidine) ligand.

In 2008, the direct hydrocarboxylation of a single alkene was reported by Rovis and co-workers, who developed a nickel-catalyzed hydrocarboxylation of styrenes with carbon dioxide.³² The reaction was conducted under very mild conditions with a nickel catalyst ($[\text{Ni}(\text{cod})_2]$ or $[\text{Ni}(\text{acac})_2]$, 10 mol%), a base additive (20 mol%), and an organozinc reagent (2.5 equiv) in tetrahydrofuran at room temperature under carbon dioxide at ambient pressure (Scheme 23). The reaction was developed for a variety of styrene analogues with electron-deficient and neutral substituents and was compatible with various functional

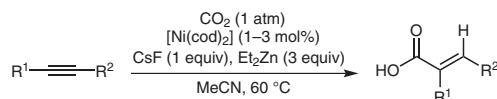
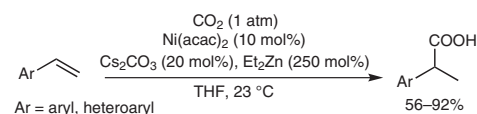
Table 7 Carboxylation with a Nickel–Bisamidine Catalyst


R ¹	R ²	Yield	Ratio a/b
Ph	H	63	>98:<2
cyclohexen-1-yl	H	67	92:8
PhCH ₂ CH ₂	H	38	47:53
Ph	Me	80	95:5
4-MeOC ₆ H ₄	Me	84	>98:<2
4-F ₃ CC ₆ H ₄	Me	84	81:19
PhCH ₂ CH ₂	Me	80	40:60
Ph	Ph	52	—

groups, such as aryl chlorides, esters, ketones, and nitriles. A single regioisomer of the corresponding α -carboxylated product was generated in good yield. The suggested mechanism involved a nickel hydride active catalyst (Scheme 24), instead of a typical Hoberg-type nickelacycle.³⁸ Insertion of styrene into the nickel–hydride bond provides benzyl nickel species **25**, a transmetalation generates the benzylic zinc species **26**, the product of net hydrozincation of the alkene,³⁹ while also regenerating ethylnickel complex **23**. β -Hydride elimination and release of ethylene from **23** generates the presumed active catalyst **24**. A separate catalytic cycle involving transmetalation back to nickel (**26** to **25**) generates another benzylic nickel species which undergoes insertion of carbon dioxide prior to transmetalation with diethylzinc, producing the hydrocarboxylation product **28** and regenerating precatalyst **23**.

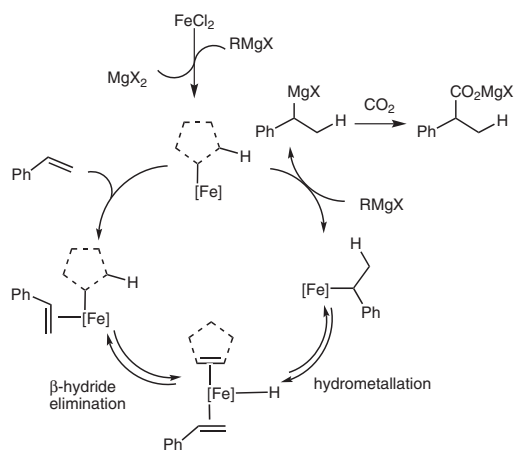
**Scheme 24****Scheme 25**

In 2011, Ma and co-workers reported a nickel-catalyzed hydrocarboxylation of alkynes with diethylzinc and carbon dioxide (Scheme 26).³³ In a typical reaction, an internal alkyne was converted into an (*E*)-2,3-disubstituted acrylic acid in the presence of [Ni(cod)₂] (1–3 mol%) as catalyst, cesium fluoride (1.0 equiv) as additive, diethylzinc (3 equiv), and carbon dioxide (1 atm). The products of a *syn* hydrocarboxylation were formed with high regio- and stereoselectivities in good to excellent yields. Symmetrical electron-rich aryl alkynes, alkyl alkynes, and unsymmetrical aryl–alkyl alkynes were found to be suitable substrates.

**Scheme 26****Scheme 23**

In the same year, Iwasawa and Takaya developed a novel catalytic hydrocarboxylation of allenes by utilizing a silyl pincer-type palladium complex as catalyst.⁴⁰ This protocol was highly attractive not only as a carbon dioxide fixation reaction but also as a methodology for the synthesis of β,γ -unsaturated carboxylic acids, owing to its wide generality and high efficiency (Scheme 25).

The alkenyl metal species generated in situ in this reaction would be less active toward carbon dioxide insertion than the benzyl nickel intermediate in the styrene hydrocarboxylation described by Rovis and co-workers.³² A different reaction pathway involving two nickel catalytic cycles was thus proposed (Scheme 27): Following initial activation of the alkyne by nickel(0) to produce **29**, carbocation leads to intermediate **30**. A β -hydride elimination and subsequent reductive elimination of ethane generates the *E*-alkenyl zinc intermediate **31** and releases nickel(0). The second cycle involves the activation of carbon dioxide by nickel(0) to form an Aresta complex,⁴¹ which then reacts with less active alkenylzinc species **32** to form the zinc carboxylate product **33** and regenerate the nickel catalyst.



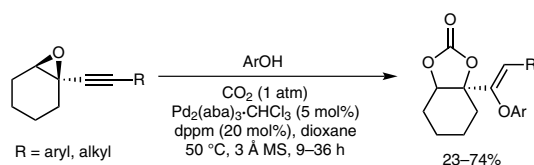
Scheme 30

In summary, the research groups of Rovis, Ma, and Tsuji have developed the methodologies for the hydrocarboxylation of alkenes and alkynes with carbon dioxide, and these reactions have twofold significance. First, these reactions demonstrate great possibilities for the use of carbon dioxide as a renewable and environmentally friendly source of carbon in organic synthesis. Further understanding of the reaction mechanisms will promote the development of new synthetic methodologies involving carbon dioxide. Second, these reactions provide versatile methods for carboxylic acid synthesis. Further improvements in these hydrocarboxylation protocols may be made, especially in terms of the use of cheap and clean reducing reagents and the expansion of their scope with respect to possible substrates.

4 Carboxylation via Oxidative Cycloaddition

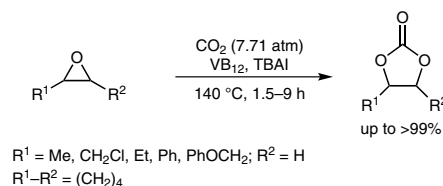
The synthesis of cyclic carbonates via cycloaddition of carbon dioxide to epoxides is one of the few processes for reducing global emissions of waste carbon dioxide and converting it into industrially useful chemical feedstocks. Au and co-workers in 2009⁴³ and North et al. in 2010⁴⁴ reviewed in detail the developments of the direct transformation of carbon dioxide into organic carbonates. Here we merely introduce a few typical examples for the synthesis of organic carbonates via oxidative cycloaddition using carbon dioxide.

In 2007, Yoshida, Ihara and co-workers developed a methodology for the synthesis of (*Z*)-phenoxyalkenyl-substituted cyclic carbonates by a palladium-catalyzed three-component coupling including a carbon dioxide fixation (Scheme 31).⁴⁵ The process produced various substituted cyclic carbonates in a highly regioselective and stereoselective manner. This reaction thus provides a new protocol for the synthesis of highly substituted cyclic carbonates in one step.



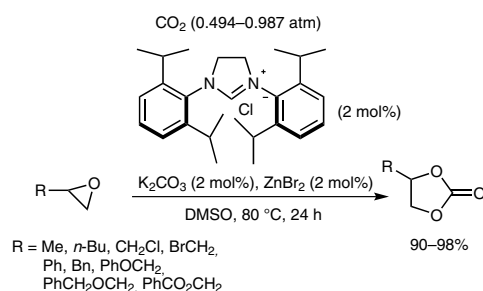
Scheme 31

In 2012, Bai et al. achieved the synthesis of cyclic carbonates from epoxides and carbon dioxide catalyzed by a cyanocobalamin (VB₁₂) and tetra(*n*-butyl)ammonium iodide system under 7.71 atm carbon dioxide pressure at 140 °C without any organic solvent (Scheme 32).⁴⁶ Propylene carbonate was obtained in >99% yield within six hours under the optimized reaction conditions. The cyanocobalamin catalyst could be recycled with water and was found to retain moderate catalytic activity.



Scheme 32

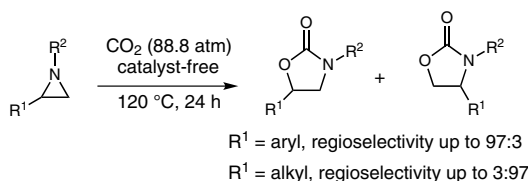
Later, Cao, Shi and co-workers developed a convenient method of coupling carbon dioxide with epoxides, catalyzed by an *N*-heterocyclic carbene and zinc bromide (Scheme 33).⁴⁷ This catalytic system exhibited excellent activity and selectivity in the cycloaddition reactions of carbon dioxide to terminal epoxides. The reactions could even be carried out under carbon dioxide pressure as low as 0.494 atm, and gave the corresponding carbonates in high yields.



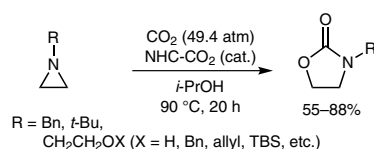
Scheme 33

In 2010, He and co-workers reported a simple approach for the facile synthesis of 5-aryl-2-oxazolidinones in excellent regioselectivity from aziridines under compressed carbon dioxide conditions in the absence of any catalyst and organic solvent (Scheme 34).⁴⁸ The reaction outcome could be tuned by subtle adjustment of the carbon dioxide pressure. The in situ formed adduct of aziridine and carbon dioxide was assumed to act as a catalyst in this reaction, which was also studied using in situ FT-IR techniques. Recently, Ikariya and co-workers described

that the imidazolium-2-carboxylates derived from N-heterocyclic carbenes and carbon dioxide served as efficient catalysts for the carbon dioxide carboxylation of tertiary aziridines bearing various substituents such as halogens, ether, olefin, ester, acetal, and nitro groups on the aziridine ring in 2-propanol, leading to 3-substituted 2-oxazolidinones in good to excellent yields (Scheme 35).⁴⁹ The NHC–CO₂ adducts facilitate nucleophilic attack of the carbon dioxide moiety on the aziridines, in which the substituents are intact during the carboxylation. The catalyst was successfully recycled up to five times with no apparent loss in activity.



Scheme 34



Scheme 35

5 Carboxylation of Carbon–Hydrogen Bonds

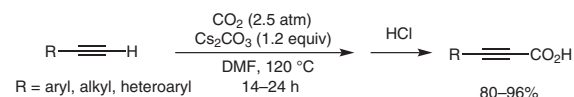
In the past decades, several interesting systems have been reported for metal-mediated reductive carboxylation of alkynes, allenes, and alkynes with carbon dioxide to form carboxylic acids or esters. However, most of those systems require either a stoichiometric amount of transition metal as reactant or an excess amount of organometallic reagent for the transmetalation process. An alternative possibility for achieving the catalytic synthesis of carboxylic acid from carbon dioxide is direct carbon–hydrogen bond activation⁵⁰ and carboxylation. Therefore, more attention has been paid in recent years to the transformation of carbon dioxide into carboxylic acids and derivatives by direct carbon–hydrogen bond activation under suitable conditions.

5.1 Carboxylation of Terminal Alkynes

Within the past years, several catalyst systems have been discovered that promote the carboxylation of terminal alkynes with formation of propiolic acids.⁵¹ This reaction concept is particularly topical in that it involves the functionalization of a carbon–hydrogen bond and the incorporation of carbon dioxide into an organic molecule without need for strong bases or aggressive organometallic reagents.

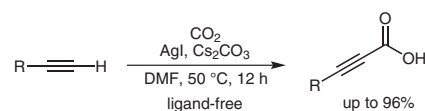
In 2011, Yu and Zhang described a simple and straightforward method for the carboxylation of terminal alkynes

with carbon dioxide (Scheme 36).⁵² The direct carbon–hydrogen bond functionalization was achieved with cesium carbonate as the base and in the absence of transition-metal catalyst, and various propiolic acids were synthesized in good to excellent yields with a wide substrate scope and good functional-group tolerance. The protocol was more atom- and step-economic with great potential for use in practical applications.

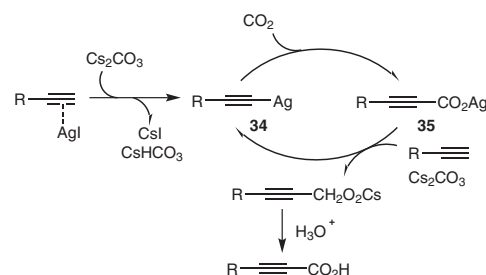


Scheme 36

In the same year, Zhang, Lu and co-workers developed a convenient approach to selectively prepare a wide range of functionalized propiolic acids by silver iodide catalyzed carboxylation of terminal alkynes using carbon dioxide as carboxylative agent under ligand-free conditions (Scheme 37).⁵³ A possible catalytic cycle for the silver(I)-catalyzed direct carboxylation of terminal alkynes with carbon dioxide was proposed as shown in Scheme 38. First, the terminal alkyne coordinates to the silver(I) salt and the acidity of the alkyne carbon–hydrogen bond is thus enhanced. Then the deprotonation reaction of the terminal alkyne by cesium carbonate eventually affords silver(I) acetylide **34**, which has been proposed as a key intermediate in the silver(I)-catalyzed alkynylation reaction⁵⁴ and silver-cocatalyzed Sonogashira reaction.^{2a} The insertion of carbon dioxide into the sp-hybridized carbon–silver bond forms silver propiolate intermediate **35**, which subsequently reacts with another terminal alkyne and cesium carbonate to release cesium propiolate, while regenerating silver(I) acetylide **34**. The acidification of the cesium propiolate after the reaction affords the propiolic acid product.



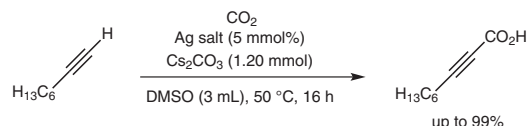
Scheme 37



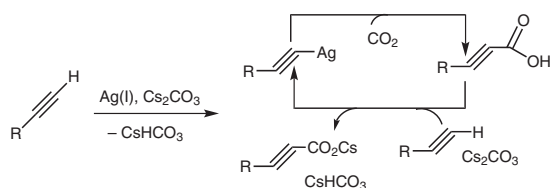
Scheme 38

The next year, Gooßen and co-workers reported a reaction protocol that permitted the C–H carboxylation of terminal alkynes at low temperatures, with mild bases, at ambient

carbon dioxide pressure, in the presence of parts-per-million quantities of an inexpensive silver catalyst (Scheme 39).⁵⁵ The carboxylation is likely to follow the catalytic cycle proposed: The reaction of silver salts or nanoparticles with a terminal alkyne affords a silver acetylide along with cesium bicarbonate. After insertion of carbon dioxide into the carbon–silver bond, cesium propiolate is released via salt metathesis with cesium carbonate, regenerating the initial silver species (Scheme 40). Further mechanistic studies of this convenient and high-yielding propiolic acid synthesis are under way.

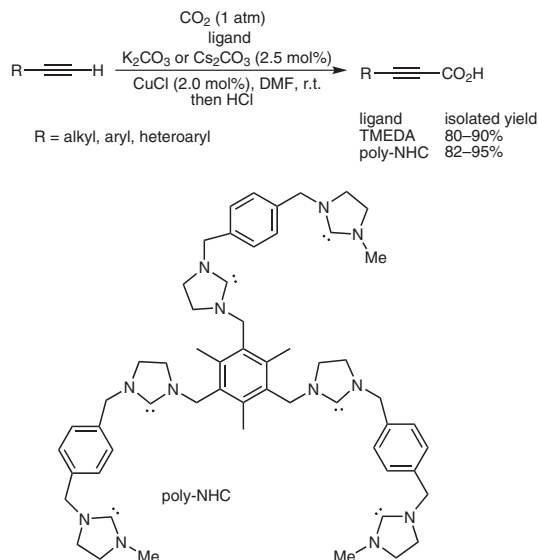


Scheme 39



Scheme 40

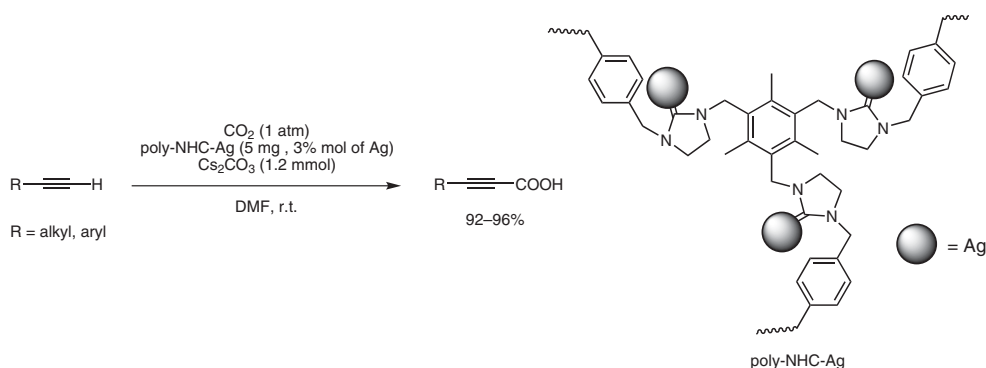
The Zhang research group developed a process where copper and a poly-NHC ligand catalyzed the transformation of carbon dioxide to carboxylic acids through carbon–hydrogen bond activation of terminal alkynes (Scheme 41).³ In this process, various propiolic acids were synthesized in good to excellent yields under ambient conditions. The most remarkable advantage of this mild reaction system is its tolerance toward a wide substrate scope. This protocol opened up access to a pool of highly functionalized propiolic acids from carbon dioxide. In addition, the poly(N-heterocyclic carbene)–copper system demonstrated the combined effect of organocatalyst and organometallic catalyst.



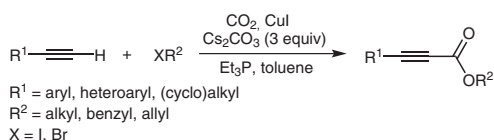
Scheme 41

In 2012, the same research group developed a poly(N-heterocyclic carbene)-supported silver nanoparticle catalyst system, and the novel nano-composite catalyst demonstrated very high activity and excellent stability and reusability in the carboxylation of terminal alkynes with carbon dioxide under ambient conditions (Scheme 42).⁵⁶ The unique N-heterocyclic carbene polymer and silver nanoparticle composite structure produced a synergistic effect on the activation of terminal alkynes and carbon dioxide that contributed to the high catalytic activity. The poly(N-heterocyclic carbene)–silver catalyst exhibited excellent substrate generality and tolerance toward various functionalities. In addition, the catalyst was stable to air and moisture and could be easily recovered and reused.

In the same year, Inamoto, Kondo and co-workers achieved an efficient coupling of terminal alkynes and carbon dioxide in the presence of alkyl halides under ambient conditions using a copper–phosphine catalyst system, providing facile access to a variety of functionalized alkyl 2-alkynoates (Scheme 43).⁵⁷



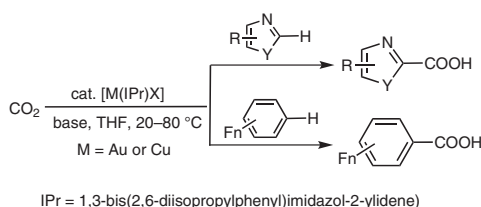
Scheme 42



Scheme 43

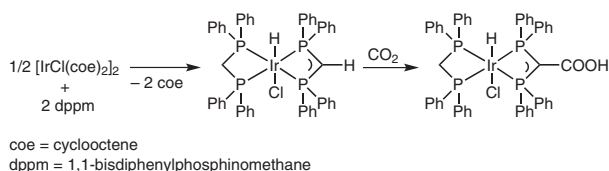
5.2 Carboxylation of sp^2 Carbon–Hydrogen Bonds

In recent years, some good results have been obtained on the direct carboxylation of sp^2 carbon–hydrogen bonds for the transformation of carbon dioxide into carboxylic acids and derivatives. Ackermann⁵⁸ and Nolan and co-workers⁵⁹ systematically reviewed the direct carboxylation of (hetero)arenes having moderately acidic sp^2 carbon–hydrogen bonds using gold and copper N-heterocyclic carbene complexes as catalysts under remarkably mild reaction conditions (Scheme 44).



Scheme 44

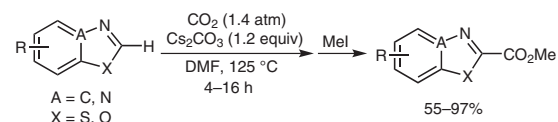
Langer et al. reported that the reaction of $[\text{Ir}_2(\mu\text{-Cl})_2(\text{coe})_4]$ with 1,1-bisdiphenylphosphinomethane under a carbon dioxide atmosphere afforded the complex $[\text{IrCl}(\text{dppm})(\text{H})\{(\text{Ph}_2\text{P})_2\text{C-COOH}\}]_2$ by initial C–H activation followed by formal insertion of carbon dioxide into the carbon–hydrogen bond of the formed diphosphanyl-methanide ligand (Scheme 45).⁶⁰



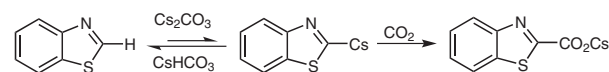
Scheme 45

Hu and co-workers described a simple and straightforward carboxylation of aromatic heterocycles such as oxazoles, thiazoles, and oxadiazoles using carbon dioxide as the one-carbon source (Scheme 46),⁶¹ and the resulting heteroaryl carboxylic acids and esters are significant compounds for medicinal and materials sciences. Remarkably, the direct C–H functionalization was possible without a metal catalyst and with only cesium carbonate as the base. Compared to other carboxylation procedures, the method was more atom- and step-economic and was practical for industrial use. The authors proposed that the carboxylation proceeds first via an uphill carbon–hydrogen bond cleavage at the C2-position, followed by car-

bon–carbon bond formation with carbon dioxide (Scheme 47).

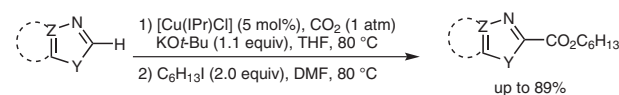


Scheme 46

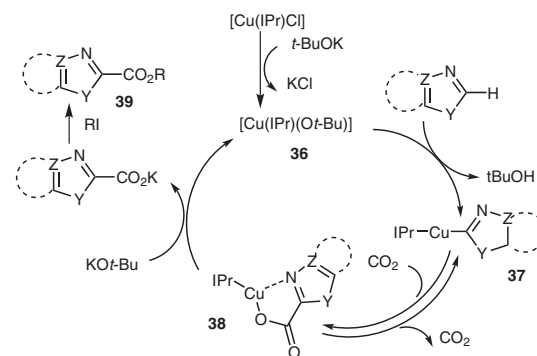


Scheme 47

In 2010, Hou and co-workers identified a novel protocol for the direct carboxylation of aromatic heterocyclic carbon–hydrogen bonds employing carbon dioxide as a one-carbon source using relatively inexpensive copper complexes as catalysts (Scheme 48).⁶² This method constituted an economical and environmentally benign process for the synthesis of heterocyclic carboxylic esters. The proposed mechanism is shown in Scheme 49. The initial metathesis reaction between $[\text{Cu(IPr)Cl}]$ and potassium *tert*-butoxide affords the copper alkoxide complex $[\text{Cu(IPr)(Ot-Bu)}]$ (**36**), which subsequently reacts with a heterocyclic compound to give the organocopper species (**37**) through activation (deprotonation) of the heterocyclic carbon–hydrogen bond. Insertion of carbon dioxide into the copper–carbon bond in **37** furnishes the carboxylate **38**, which could release carbon dioxide to regenerate **37**. Further reaction of **38** with potassium *tert*-butoxide would regenerate the copper alkoxide **36** and release the potassium carboxylate, which after reaction with an alkyl iodide affords the ester product **39**.



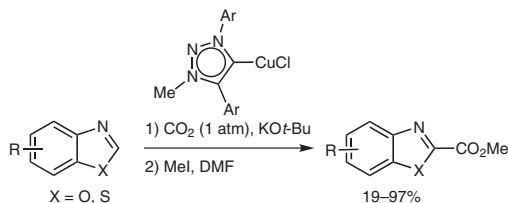
Scheme 48



Scheme 49

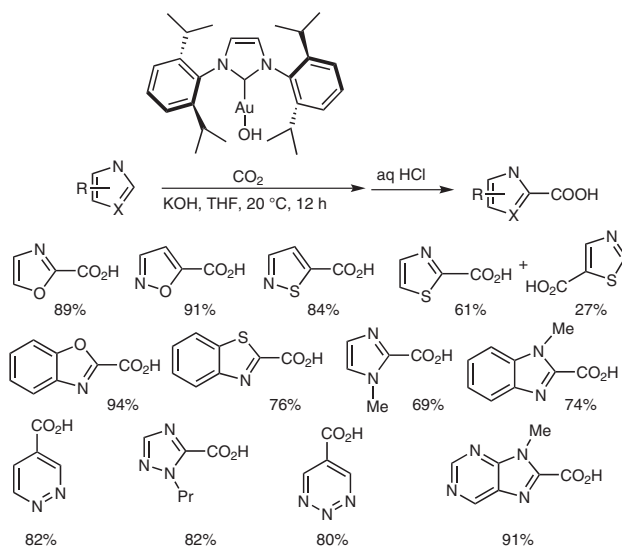
Later, the same research group also reported the direct C–H carboxylation of benzoxazole and benzothiazole derivatives with carbon dioxide to give the corresponding es-

ters in excellent yields after treatment with alkyl iodide using 1,2,3-triazol-5-ylidene-copper(I) complexes (tzNHC-Cu) as catalysts (Scheme 50).⁶³ The tzNHC-copper(I) complex [(TPr)CuCl] worked somewhat more effectively than the corresponding imidazol-2-ylidene-copper(I) complex [(IPr)CuCl] to give the carboxylation product in higher yields.



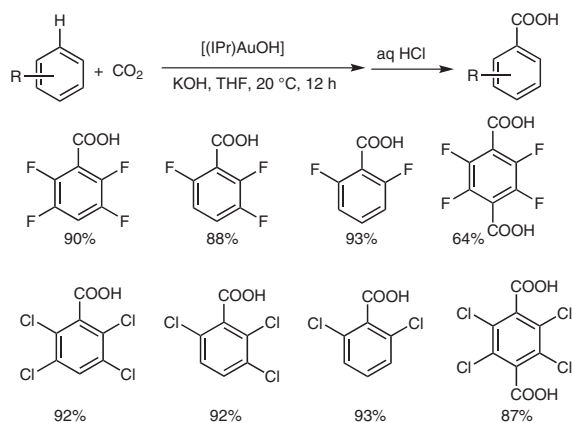
Scheme 50

Nolan and Boogaerts described that NHC-gold(I) hydroxide complexes could catalyze the carboxylation of heterocycles (Scheme 51) and carbocycles (Scheme 52) with high regioselectivity for the most acidic C-H position.⁶⁴ The selective carboxylation could be rationalized in terms of simple acid/base theory. The proposed catalytic intermediates have been isolated, characterized, and shown to be active in the carboxylation reaction (Scheme 53).

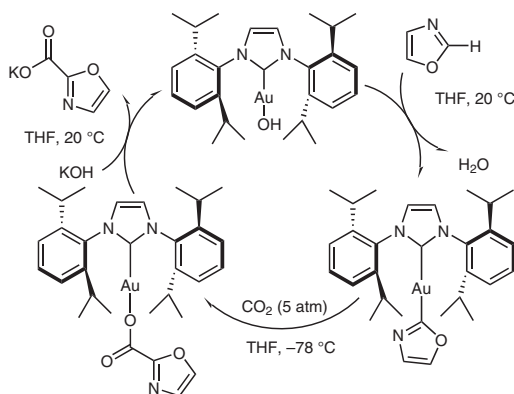


Scheme 51

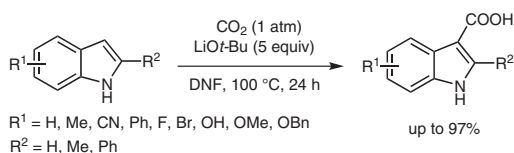
In 2012, Kobayashi and co-workers discovered a simple and straightforward method for the preparation of indole-3-carboxylic acids through the direct carboxylation of indoles with atmospheric pressure of carbon dioxide under basic conditions (Scheme 54).⁶⁵ The relatively large excess of base used, and the careful choice thereof, were key to achieving the carboxylation of indoles in excellent yields. Although the precise mechanism for the lithium *tert*-butoxide mediated carboxylation of indole derivatives has not been elucidated, a tentative mechanism was proposed (Scheme 55).



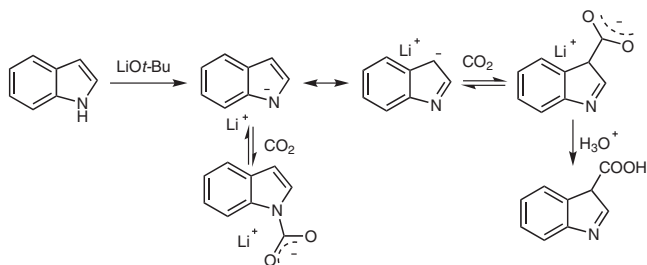
Scheme 52



Scheme 53



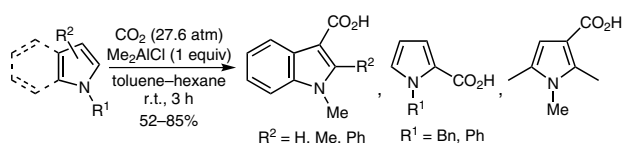
Scheme 54



Scheme 55

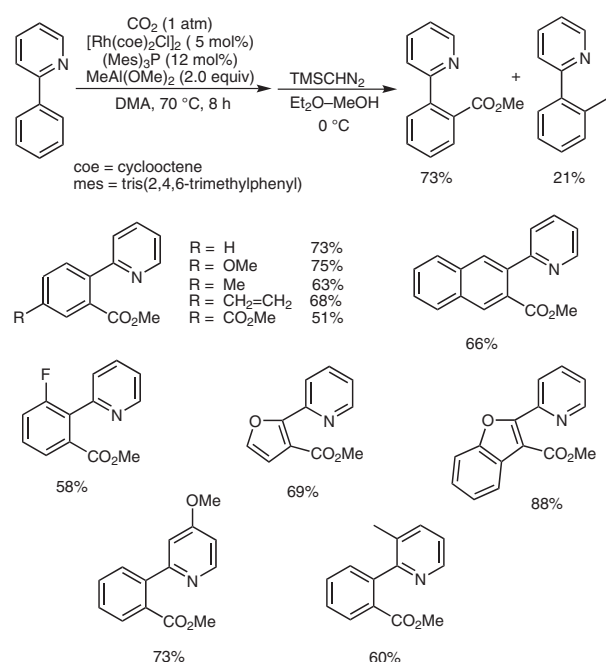
Hattori et al. were successful in applying the Lewis acid mediated carboxylation of arenes with carbon dioxide to 1-substituted indoles and pyrroles by using dialkylaluminum chlorides instead of aluminum trihalides (Scheme 56).⁶⁶ The carboxylation of 1-methylindoles and 1-benzyl- and 1-phenylpyrroles proceeded regioselectively

with the aid of an equimolar amount of Me_2AlCl under carbon dioxide (27.6 atm) at room temperature to afford the corresponding indole-3-carboxylic acids and pyrrole-2-carboxylic acids in 61–85% yields, while the same treatment of 1,2,5-trimethylpyrrole afforded the 3-carboxylic acid in 52% yield.



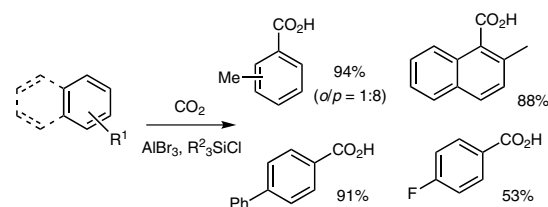
Scheme 56

In 2011, Iwasawa and co-workers succeeded in developing a catalytic direct carboxylation of aromatic compounds via chelation-assisted carbon–hydrogen bond activation (Scheme 57).⁶⁷ Various substituted and functionalized 2-arylpyridines and 1-arylpyrazoles underwent the carboxylation in the presence of the rhodium catalyst and a stoichiometric methylating reagent, dimethyl(methoxy)aluminum, to give carboxylated products in good yields. By carrying out the reaction in the presence of methylaluminum reagent, nucleophilic carbon–carbon bond formation was achieved with carbon–hydrogen bond activation. This approach demonstrates a promising application of the carbon–hydrogen bond activation strategy in the field of carbon dioxide fixation.



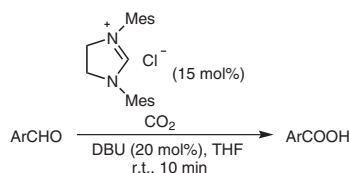
Scheme 57

In 2010, Hattori et al. reported the direct carboxylation of arenes and halobenzenes with carbon dioxide by the combined use of aluminum tribromide and tri(alkyl or aryl)silyl chloride (Scheme 58).⁶⁸ The Lewis acid mediated direct carboxylation of aromatic compounds with carbon dioxide was efficiently promoted by the addition of silyl chlorides bearing three alkyl and/or aryl substituents in total on the silicon atom. Toluene, xylenes, mesitylene, and some other alkylbenzenes were treated, neat, with a 1:1 mixture of aluminum tribromide and triphenylsilyl chloride under carbon dioxide (27.6 atm) at room temperature to give the corresponding carboxylic acids in 60–97% yields, based on aluminum tribromide. Polycyclic arenes, including naphthalene, phenanthrene, and biphenyl, were regioselectively carboxylated in 91–98% yields with the aid of 1 mole equivalent of aluminum tribromide and triphenylsilyl chloride in an appropriate solvent, chosen from benzene, chlorobenzene, and fluorobenzene. These solvents, as well as bromobenzene, resist carboxylation; they are carboxylated in moderate yields, however, when treated with a 1:5 mixture of aluminum tribromide and triisopropylsilyl chloride at elevated temperatures.

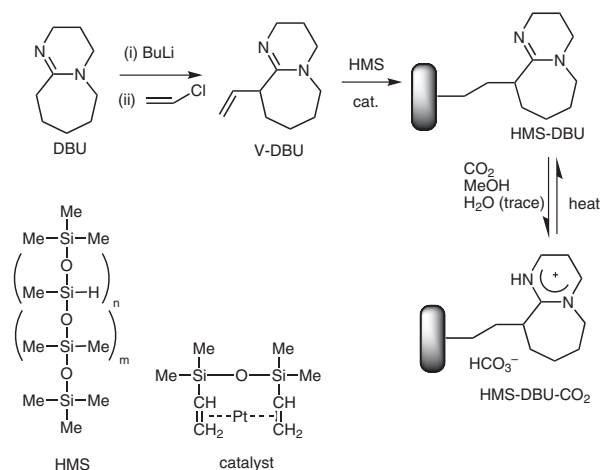


Scheme 58

Nair et al. described a facile NHC-mediated reaction of aromatic aldehydes with carbon dioxide leading to carboxylic acids (Scheme 59).⁶⁹ This protocol was mechanistically important, and could serve as a tool for the sequestration of carbon dioxide. The following year, Beckmana and Munshi reported the synthesis and use of methylhydrosiloxane dimethyl siloxane copolymer (trimethylsiloxane-terminated) (HMS)-supported DBU, known as HMS-DBU, which absorbs carbon dioxide almost 100 mol% with respect to the available functional groups (Scheme 60).⁷⁰ The material obtained after the reaction with carbon dioxide is denoted as HMS-DBU- CO_2 , which acts as a reversible carbon dioxide carrier (RCC) that has been used to perform the carboxylation of ketones to β -keto esters under ambient carbon dioxide pressure and temperature (Table 9). The extent of carboxylation might be correlated to the enol content of the substrate. Usually the reactions involving carbon dioxide were carried out at a very high pressure, which might not be economically suitable. However, this work described a method wherein a low pressure of carbon dioxide could be conveniently used for chemical transformations, thus making it more economic.



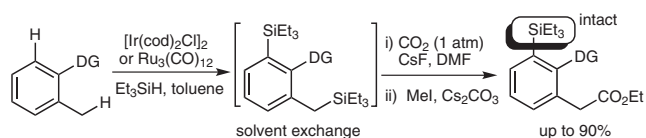
Scheme 59



Scheme 60

5.3 Carboxylation of sp^3 Carbon–Hydrogen Bonds

One of the most challenging transformations in current organic chemistry is the catalytic carboxylation of a $\text{C}(\text{sp}^3)\text{--H}$ bond using carbon dioxide gas, an inexpensive and ubiquitous one-carbon source. In 2012, Mita, Sato and Michigami established a sequential protocol for $\text{C}(\text{sp}^3)\text{--H}$ carboxylation by employing a nitrogen-directed, metal-assisted, C--H activation and catalytic silylation reaction in conjunction with fluoride-mediated carboxylation with carbon dioxide (Scheme 61).⁷¹ The carboxylation proceeded only at the benzylic $\text{C}(\text{sp}^3)\text{--Si}$ bond, not at the aromatic $\text{C}(\text{sp}^2)\text{--Si}$, and was thus advantageous for further manipulations of the products.

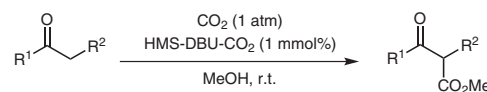


Scheme 61

6 Electrochemical Carboxylation

Electro-organic synthesis is becoming an important green synthetic method, as electrons are used as the reagents thereby avoiding the use of other toxic reagents. In addition, pure products can be obtained with high selectivity by electrochemical synthesis, and this is certainly an advantage to industry. Electrochemical fixation of carbon

Table 9 Carboxylation of Ketones to β -Keto Esters Using HMS-DBU- CO_2

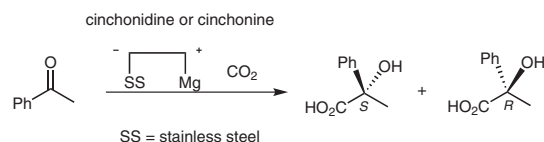


Substrate	Product	Yield (%)
		99
		99
		32
		99
		96
		98
		90
		29
		18
		12
		9

dioxide is a useful and attractive method for efficient synthesis of various carboxylic acids and derivatives.

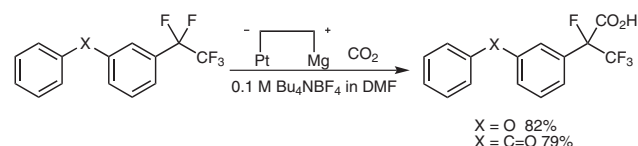
In 2009, Lu and co-workers developed a method for the selective fixation of carbon dioxide by way of asymmetric electrochemical carboxylation of prochiral acetophenone (Scheme 62).⁷² In an undivided cell, the pharmaceutically active intermediate 2-hydroxy-2-phenylpropionic acid (atrolactic acid) was produced from prochiral acetophenone in the presence of two kinds of chiral alkaloids, cinchonidine and cinchonine, acting as the inductors which were inclined to afford *R*- and *S*-configured products, re-

spectively. Since the alkaloid has a strong tendency to adsorb to the surface of the cathode, three different cathode materials (stainless steel, platinum and copper) were applied. Eventually, very distinct results were obtained. When the stainless steel was used as the cathode, the highest enantiomeric excess (29.8%) was achieved with an electrocarboxylation yield of 24.5%. Using cinchonidine and cinchonine as the inductors, the ee value of the desired 2-hydroxy-2-phenylpropionic acid was also measured as a function of the concentration ratio of the alkaloid to the cocatalyst of butanol, supporting electrolyte, temperature, charge passed, current density and solvent.

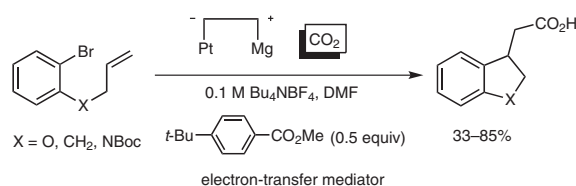


Scheme 62

The same year, Senboku et al. achieved the synthesis of 2,3,3,3-tetrafluoro-2-(3-phenoxyphenyl)propanoic acid and 2-(3-benzoylphenyl)-2,3,3,3-tetrafluoropropanoic acid (tetrafluorinated fenoprofen and ketoprofen) by electrochemical carboxylation of pentafluoroethylarenes (Scheme 63).⁷³ Two years later, the same research group developed the facile synthesis of 2,3-dihydrobenzofuran-3-ylacetic acids and related analogues by a novel electrochemical aryl radical generation and its 5-*exo* cyclization followed by a carboxylation sequence of 2-allyloxybromobenzenes by using methyl 4-*tert*-butylbenzoate as an electron-transfer mediator (Scheme 64).⁷⁴



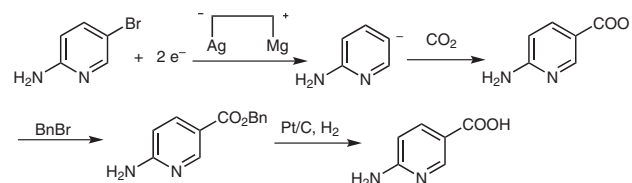
Scheme 63



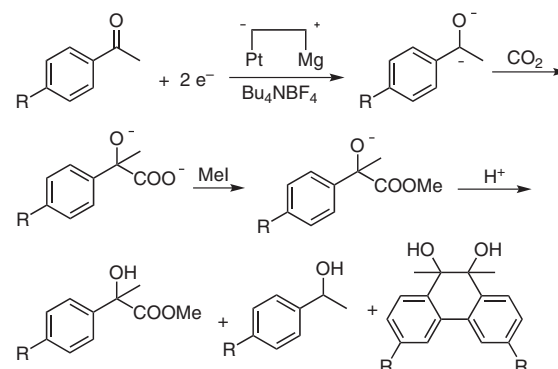
Scheme 64

In 2010, Huang and co-workers investigated a new electrochemical procedure for the electrocatalytic carboxylation of 2-amino-5-bromopyridine with carbon dioxide in the ionic liquid 1-butyl-3-methylimidazolium tetrafluoroborate (BMIMBF₄) to produce 6-aminonicotinic acid (Scheme 65).⁷⁵ 6-Aminonicotinic acid was obtained in 75% yield and 100% selectivity under the optimized conditions. Moreover, the ionic liquid was successfully recycled.

Later, the authors reported a procedure for the electrocatalytic carboxylation of aromatic ketones with carbon dioxide in the same ionic liquid, BMIMBF₄, to provide α -hydroxycarboxylic acid methyl esters in moderate yield (Scheme 66).⁷⁶ The results showed that the yields were strongly affected by various factors: temperature, current density, charge passed, electrode material and substrate concentration. The ionic liquid was recyclable.

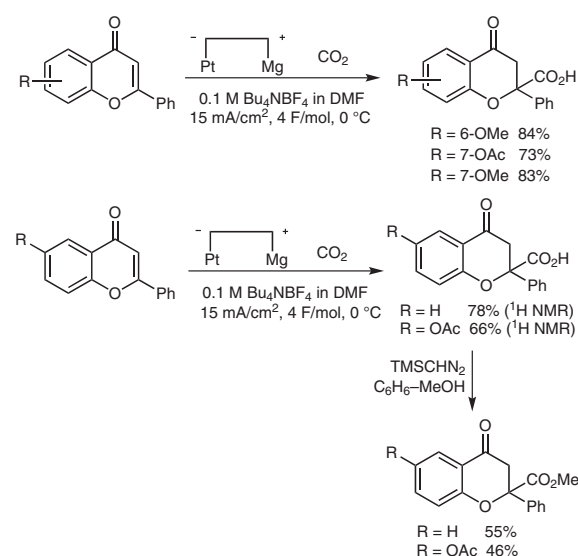


Scheme 65



Scheme 66

In 2012, Senboku et al. successfully carried out the electrochemical carboxylation of several flavones by using a one-compartment cell equipped with a platinum cathode and a magnesium anode in *N,N*-dimethylformamide to afford flavanone-2-carboxylic acids in moderate to good yields (Scheme 67).⁷⁷ They also carried out some mecha-



Scheme 67

nistic studies and proposed a reaction mechanism to account for the exclusive mono-carboxylation of flavones at the C2-position.

7 Conclusion

Within the past several years, tremendous achievements have been realized in the catalytic transformation of carbon dioxide into value-added fine chemicals. For example, the insertion of carbon dioxide into metal–element bonds is already a well-established procedure, and appears to be interesting for special organic products for fine chemical synthesis. In the carboxylation of unsaturated organic compounds, the oxidative cycloaddition catalyzed by low-valent-transition-metal complexes is an efficient and general approach by which a series of unsaturated carboxylate derivatives can be produced. The dimethyl carbonate synthesis and alternating copolymerization with epoxides are one step away from commercialization. Electrochemical reduction of carbon dioxide alone results in the formation of oxalic acid and its derivatives such as glycolic and glyoxylic acids. By the electrochemical reactions of alkenes, ketones, or organic halides in the presence of carbon dioxide, carboxylic groups are introduced to organic substrates.

The use of milder conditions, such as room temperature or low carbon dioxide pressure, for the transformation of carbon dioxide into useful chemicals remains a great challenge for organic synthetic chemists. The milder conditions would likely alleviate the problem of limited substrate scope as well as allow more widespread use of these methods, including applications in large-scale synthesis. The manipulation of moisture- and air-sensitive reagents also limits the methodology.

It is noteworthy that the ideal direct C–H carboxylation with carbon dioxide has been achieved, which presents a challenging and promising goal in the combination of both C–H activation and transformation of carbon dioxide. The ideal direct carboxylation of hydrocarbons might require a new strategy to reach such goals. The direct carboxylation of (hetero)arenes has been accomplished under remarkably mild reaction conditions through the use of highly basic gold- or copper–NHC complexes, in combination with a stoichiometric potassium or cesium base. While these findings constitute an important breakthrough in the area of carbon dioxide fixation, a limitation of these protocols is their restriction to moderately acidic carbon–hydrogen bonds. Therefore, additional challenges include the direct carboxylation of less acidic substrates and unactivated sp³ carbon–hydrogen bonds, as well as the use of milder stoichiometric bases.

To meet the economic and environmental requirements, and in the meantime offer ideal carbon sources for the production of useful organic compounds, more attention should be paid to the elucidation of the reaction mechanism at the molecular level to further develop highly efficient catalytic systems. Considering the importance of

carbon dioxide utilization, it can be anticipated that more exciting developments are to come in this research area.

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